Environ. Sci. Technol. **2010,** *44,* 688–693

Evidence of Compound-Dependent Hydrodynamic and Mechanical Transverse Dispersion by Multitracer Laboratory Experiments

GABRIELE CHIOGNA, CHRISTINA EBERHARDT, PETER GRATHWOHL, OLAF A. CIRPKA, AND MASSIMO ROLLE*

Center for Applied Geosciences, University of Tübingen, Sigwartstrasse 10, D-72076 Tübingen, Germany

Received August 6, 2009. Revised manuscript received November 24, 2009. Accepted December 1, 2009.

Mass transfer, mixing, and therefore reaction rates during transport of solutes in porous media strongly depend on dispersion and diffusion. In particular, transverse mixing is a significant mechanism controlling natural attenuation of contaminant plumes in groundwater. The aim of the present study is to gain a deeper understanding of vertical transverse dispersive mixing of reaction partners in saturated porous media. Multitracer laboratory experiments in a quasi two-dimensional tank filled with glass beads were conducted and transverse dispersion coefficients were determined from high-resolution vertical concentration profiles. We investigated the behavior of conservative tracers (i.e., fluorescein, dissolved oxygen, and bromide), with different aqueous diffusion coefficients, in a range of grain-related Péclet numbers between 1 and 562. The experimental results do not agree with the classical linear parametric model of hydrodynamic dispersion, in which the transverse component is approximated as the sum of pore diffusion and a compound-independent mechanical dispersion term. The outcome of the multitracer experiments clearly indicates a nonlinear relation between the dispersion coefficient and the average linear velocity. More importantly, we show that transverse mechanical dispersion depends on the diffusion coefficient of the compound, at least at the experimental bench-scale. This result has to be considered in reactive-transport models, because the typical assumption that two reactants with different aqueous diffusive properties are characterized by the same dispersive behavior does not hold anymore.

1. Introduction

Dispersion in porous media is a process of utmost importance in many scientific disciplines and technical applications, including contaminant hydrology, water treatment, oil reservoir engineering, and chemical engineering. Hydrodynamic dispersion can be described as the sum of two simultaneous contributions: pore diffusion, resulting from Brownian motion, and mechanical dispersion, resulting from variations of local velocities in the porous medium, both in magnitude and direction (*1*).

The efforts to describe dispersion of solutes in uniform porous media began more than 100 years ago (*2*), with major developments in the 1960s and 1970s. An overview of the principal dispersion theories is given in the textbooks of Bear (*1, 3*). Two main approaches to hydrodynamic dispersion can be distinguished: exact mathematical methods which solve the governing flow and transport equations for some simplified boundary conditions (*4*) and statistical methods, for which the description of transport through a porous medium is based on probability distributions and averaging procedures to obtain macroscopic properties (e.g., refs *5*−*9*). The development of these statistical methods led to the description of dispersion as a tensorial property (*6, 9*). For a Cartesian coordinate system oriented along the principal flow direction, the dispersion tensor can be simplified to a second-order diagonal tensor in which components in the longitudinal (parallel to the flow) and transverse (perpendicular to the flow) directions can be distinguished.

The transverse dispersion components, typically smaller than the longitudinal ones, are very important for transverse mixing, which is a process of major interest for the spreading and mixing of contaminant plumes in groundwater (e.g., refs *10*−*12*). Since reactions take place at the microscopic (pore) scale, pore-scale dispersion was identified to be a critical process for overall contaminant degradation (*13*). Therefore, many recent studies addressed the topic of transverse dispersive mixing using both experimental and modeling approaches. At the laboratory scale, flow-through experiments have been performed to investigate mixing and mixing-controlled abiotic and microbially mediated reactions in porous media (e.g., refs *11, 14*−*16*). At the field scale, high-resolution site investigations (*17*) showed the importance of mixing-controlled reactions occurring at the narrow (mm−cm) fringes of organic contaminant plumes. Mathematical modeling studies using both numerical (e.g., refs *10, 18*) and analytical techniques (e.g., refs *12, 19, 20.*) focused on transport and mixing-controlled reactions. At the pore scale, microfluidic experiments (*21*) and pore-scale simulations (e.g., refs *22, 23*) were recently carried out to investigate mixing for well-defined porous media geometries.

Reactions in porous media require mixing of the reaction partners as well as pore-scale transport of the reaction product(s). Depending on reaction pathways, a multitude of different compounds and species undergo dispersion processes. Thus the goal of this study is to investigate the dispersive behavior of different compounds by performing multitracer laboratory flow-through experiments. Conservative tracer tests were carried out at different velocities with the simultaneous continuous injection of at least two compounds in a homogeneous isotropic porous medium. We chose this simplified experimental set up to reduce the complexity of the problem and to gain insights into the fundamental physics that govern transverse dispersion of solutes in saturated porous media. An improvement in the description of the diffusive/dispersive behavior of different compounds is of primary importance to quantify and understand mixing-controlled reactions (e.g., in groundwater or in chemical flow-through reactors), which require the simultaneous presence of the reactive partners at the location where the reaction occurs.

2. Theoretical Background

Hydrodynamic dispersion coefficients have been used to describe dispersion of solutes in porous media. These coefficients are defined as the additive contribution of a pore diffusion coefficient, D_p $[L^2T^{-1}]$, and a mechanical dispersion

* Corresponding author e-mail: massimo.rolle@uni-tuebingen.de; tel: +49 70712973171; fax: +49 7071295059.

10.1021/es9023964

© 2010 American Chemical Society

Published on Web 12/18/2009

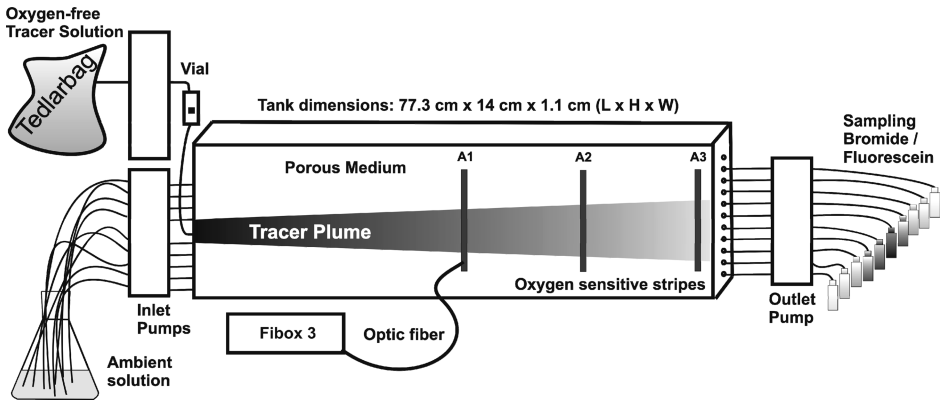

FIGURE 1. Experimental setup.

coefficient, D_{mech} $[L^2T^{-1}]$. Therefore the coefficient of hydrodynamic dispersion in transverse direction D_t can be written as:

$$D_t = D_p + D_{mech} \tag{1}$$

However, as already pointed out by Bear (*1*), a clear separation of the two contributions is rather artificial since hydrodynamic dispersion includes both processes in an inseparable form.

Velocity ranges in which pore diffusion or mechanical dispersion is dominating are commonly distinguished (*1, 23*) by the value of the dimensionless Péclet number (*Pe*):

$$Pe = \frac{vd}{D_{aq}} \tag{2}$$

in which v $[LT^{-1}]$is the flow velocity in the porous medium, d $[L]$ is the length of a pore channel, represented by the average grain size (diameter), and D_{aq} $[L^2T^{-1}]$ is the diffusion coefficient of the compound in water.

In porous media, molecular diffusion of solutes can only occur in the pore space. Considering the diminished cross-sectional area available for diffusion, the tortuous nature of the pores, and the size of the pores, the pore diffusion coefficient is smaller than the free aqueous diffusion coefficient. In most practical cases, the tortuosity and pore size distribution are unknown and the pore diffusion coefficient is derived from empirical correlations as a function of the aqueous diffusion of the solute, and the porosity n [-] of the medium (*24, 25*). Under water-saturated conditions D_p can be approximated as:

$$D_p \approx D_{aq}n \tag{3}$$

Mechanical dispersion, considered as a pore-scale process, quantifies the scattering of the particles in the porous medium due to the presence of the grains and the fluctuations about the average velocity of the fluid. Conceptualizing random walks in a network of capillaries, the transverse mechanical dispersion was defined as a quantity linearly proportional to the velocity and to the grain size (e.g., refs *7, 26*). Nowadays it is commonly accepted (e.g., 27) to define this term with the linear parametric model proposed by Scheidegger (*9*):

$$D_{mech} = \alpha_t v \tag{4}$$

where the transverse dispersivity α_t $[L]$ is a property of the porous medium assumed proportional to the grain size under some restrictive assumptions (*28*). If we accept this description, it seems that we can treat molecular diffusion and mechanical dispersion as two separate phenomena, that is, D_{mech} is identical for all compounds. Consequently two different compounds will show the same dispersive behavior at high velocities, where the contribution of pore diffusion to the overall dispersion is negligible.

Based on a statistical approach with spatial averaging on a representative elementary volume (REV), Bear and Bachmat (*5*) theoretically derived the following expression for the transverse component of the dispersion tensor:

$$D_t = D_p + \alpha_t v f(Pe, \delta) \tag{5}$$

where $f(Pe,\delta)$ is defined as:

$$f(Pe, \delta) = \frac{Pe}{Pe + 2 + 4 \cdot \delta^2} \tag{6}$$

$$\delta = \frac{d}{a} \tag{7}$$

in which d $[L]$ is the length of a pore channel and a $[L]$ its hydraulic radius. The value of the ratio δ can range between 4 and 20 (*8, 29*). The introduction of the function $f(Pe,\delta)$ was justified by physical considerations about the relative rate of growth of the volume occupied by a moving elementary mass of solute in a porous medium. This function implies a compound-dependency of the mechanical dispersion term. Bear and Bachmat also considered the possibility of introducing a noninteger exponent to reproduce nonlinear effects observed in experimental studies (e.g., refs *16, 29*−*35*). This approach clearly points out that molecular diffusion is a key process inseparable from mechanical dispersion; only diffusion allows transfer of molecules from a streamline to another.

The parametric model of eq 5 allows overcoming inconsistencies of eq 4 with experimental results. Several recent studies clearly indicate the nonlinearities (*16, 23, 29*−*35*) and temperature dependencies (*34*) of the transverse dispersion over large velocity ranges and at large values of *Pe*. Therefore, this study focuses on well-controlled experiments to elucidate the compound dependency of mechanical transverse dispersion as a function of flow velocity. To our knowledge this is the first study specifically dedicated to this phenomenon.

3. Material and Methods

The experimental setup consists of a quasi two-dimensional flow-through system, made of glass, Teflon, and alumina, with inner dimensions 77.3 cm × 14.0 cm × 1.1 cm (L × H × W). The tank is equipped with 10 equally spaced (1.2 cm) inlet and outlet ports connected to three high-precision peristaltic pumps (IPC-N, Ismatec, Glattburg, Switzerland), which allow maintaining constant flow boundary conditions, as illustrated in Figure 1. We used Tygon tubes for all ports, except the inlet port of the tracer where a Fluran tube was used to minimize atmospheric oxygen diffusion into the tube. Similar setups were recently used for flow-through experiments with conservative and reactive tracers (e.g., ref (*33*))

and to study coupled transport and biodegradation processes (e.g., refs *14, 15*).

Oxygen concentration gradients were measured at three oxygen-sensitive polymer stripes (10 cm × 0.5 cm, PreSens, Germany) attached to the inner wall of the tank (Figure 1), at a distance of 40.5 cm (A1), 57 cm (A2), and 74 cm (A3) from the inlet.

Before the injection into the tank, the tracer solution passed through a vial equipped with an oxygen-sensitive membrane spot to check the inlet oxygen concentration.

The tank was homogeneously filled with glass beads with diameters ranging between 0.5 and 0.75 mm (Fisher Scientific GmbH, Germany). During the filling procedure, the water level was always maintained above the height of the glass beads to avoid trapping of air into the porous medium. In the experiments the water table was kept below the surface of the porous medium, thus maintaining an unsaturated zone on top of the aquifer to avoid a shortcut of the tracer along a free water surface. The porosity of the system was determined from evaluation of breakthrough curves of the injected solutes, obtaining an average value of 0.39.

We performed tracer tests with a bromide solution (42.7 or 20 mg/L) depleted in oxygen and a sodium fluorescein solution (20 mg/L) depleted in oxygen at different seepage velocities. The oxygen depletion was performed by stripping the solutions with nitrogen gas and then storing them in a gastight Tedlar bag (Alltech, Germany). Hence, three different tracers (i.e., fluorescein, bromide, and oxygen) with characteristic aqueous diffusion coefficients were selected. Fluorescein is often used as conservative tracer and has, at 22 °C, an aqueous diffusion coefficient (D_{aq}) of 4.8×10^{-10} m^2/s (determined after Worch (*36*)). This value is similar to those of many organic pollutants frequently found in contaminated groundwater. Bromide and oxygen are characterized by considerably higher aqueous diffusion coefficients of 1.96×10^{-9} m^2/s and 1.97×10^{-9} m^2/s at 22 °C, respectively (*36*−*38*).

The tracer solution was injected into the middle port (port 6, counted from the bottom of the tank). The other 9 ports were used for the injection of the ambient solution: tap water when we applied fluorescein as tracer and Millipore water in case of bromide. The oxygen concentration in those ambient solutions was in equilibrium with the atmosphere.

After steady-state conditions were reached (i.e., after the exchange of at least two pore volumes) water samples for fluorescein or bromide measurements were taken at the outlet ports (Figure 1) and analyzed by UV-spectrometry and ion-chromatography (Dionex DX-120, Germany), respectively. The oxygen concentrations were measured at 45 points along the 3 oxygen-sensitive polymer stripes from outside with a noninvasive optode technique using an optical fiber as a light guide between the oxygen meter (Fibox 3, PreSens, Germany) and the sensor foil (Figure 1). The principle of the sensor operation is based on quenching of luminescence caused by collision between molecular oxygen and luminescent dye molecules in the excited state. The oxygen sensing is a luminescence lifetime detector. The detection limit is at 0.015 mg/L dissolved oxygen.

The injected solutions containing fluorescein and bromide were depleted in oxygen and were injected separately for each selected flow velocity. In this way, two compounds (fluorescein and oxygen, or bromide and oxygen, respectively) could always be measured during the same experimental run at a given flow velocity. The good agreement between the measured value of the transverse dispersion coefficient for oxygen during the runs with the oxygen/fluorescein solution and the oxygen/bromide solution guarantees the same hydraulic conditions of the system. The fact that the plume simultaneously carried two compounds allowed studying the effect of molecular diffusion on transverse dispersion without disturbance by nonidentical experimental conditions (flow rates, packing, etc.) which may occur in individual tests with only one compound. The relative experimental error for the determined transverse dispersion coefficient can be estimated in the order of 15%. Further details about the measurements of each compound are reported in the Supporting Information.

The determination of the transverse dispersion coefficients was performed by applying the analytical solution of the 2-D steady-state advection−dispersion equation neglecting longitudinal dispersion:

$$v\frac{\partial C}{\partial x} - D_t\frac{\partial^2 C}{\partial z^2} = 0 \tag{8}$$

in which C is the concentration $[ML^{-3}]$, x $[L]$ is the coordinate in the direction of the flow, and z $[L]$ is the transverse coordinate. This equation is applicable, as pointed out in ref *39*, since longitudinal dispersion is negligible under steady-state conditions at a sufficient distance from the source.

As boundary condition, we consider a line source determined by the continuous injections of two constant concentration solutions:

$$C(x = 0, z) = \begin{cases} C_0 \ \forall |z| < \dfrac{W}{2} \\ C_1 \ \forall |z| > \dfrac{W}{2} \end{cases} \tag{9}$$

where C_1 is the uniform ambient concentration of the compound, C_0 is the concentration in the injected tracer solution, and W [L] is the source width (1.2 cm in our experimental setup).

In a semi-infinite domain, the general solution to the problem is (adapted after ref *40*):

$$C(x, z) = C_1 + \frac{C_0 - C_1}{2}\left[\mathrm{erf}\left(\frac{z + \frac{W}{2}}{2\sqrt{D_t\frac{x}{v}}}\right) - \mathrm{erf}\left(\frac{z - \frac{W}{2}}{2\sqrt{D_t\frac{x}{v}}}\right)\right] \tag{10}$$

This analytical solution was fitted to the measured data points using the Nelder−Mead simplex algorithm in which the dispersion coefficient D_t and a shift a, which accounts for the usually small deviation of the plume centerline from the injection port, were the fitting parameters.

4. Results and Discussion

Concentration profiles of the conservative tracers at different flow velocities (1, 3.5, 9.5, and 16.5 m/d) are shown in Figure 2. To compare the results for the three different tracers and to eliminate systematic errors due to the different measurement techniques the data are plotted as normalized concentrations, defined as $(C - C_1)/(C_0 - C_1)$, along the vertical cross-section. Measured solute concentrations at the outlet ports (fluorescein and bromide) and at the last oxygen sensitive stripe A3 (oxygen), respectively, are shown by markers, whereas the lines indicate the fitted analytical solution. A remarkable difference in the physical displacement of fluorescein and the compounds with larger aqueous diffusion coefficient (bromide and oxygen) can be clearly noticed for all seepage velocities. The role played by D_{aq} is obvious at low flow velocities, where the dispersion process is dominated by diffusion, but the difference is still relevant at higher flow velocities, as can be clearly seen from the difference in the peak concentration. A higher value of the dispersion coefficient leads to a broader Gaussian concentration profile with lower peak concentration at the same flow velocity.

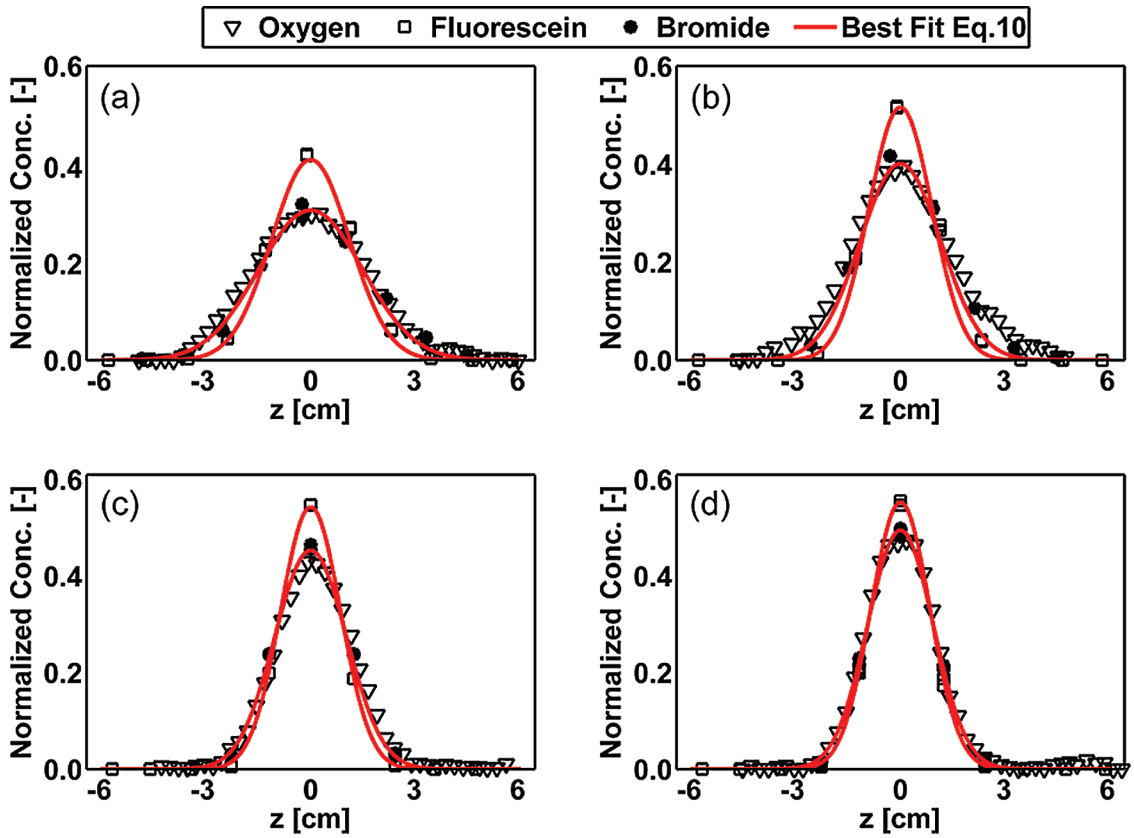

FIGURE 2. Comparison of vertical concentration profiles for oxygen, fluorescein, and bromide at flow velocities of 1 m/d (a), 3.5 m/d (b), 9.5 m/d (c), and 16.5 m/d (d). Solid lines represent the best fit obtained with eq 10 for the different compounds.

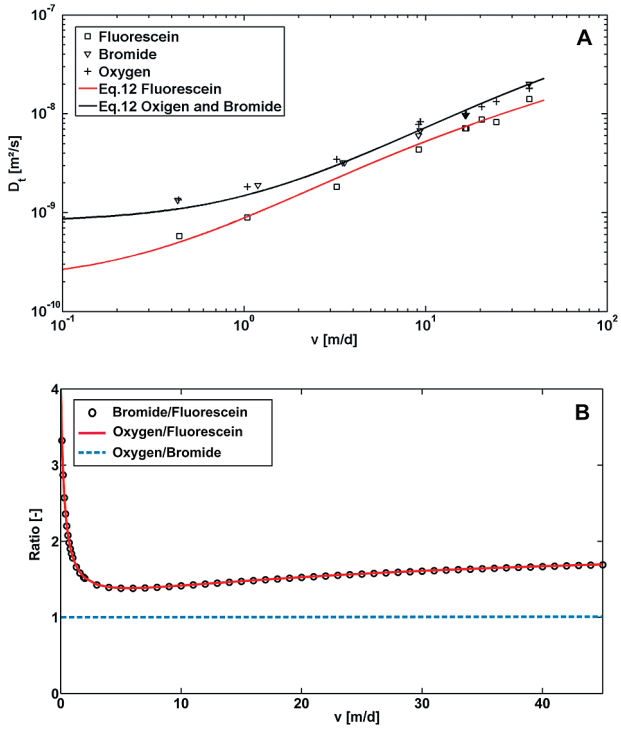

FIGURE 3. (A) Transverse dispersion coefficients for fluorescein, bromide, and oxygen as a function of the seepage velocity; lines represent the non linear parametric model (eq 12), in which bromide and oxygen (black lines) almost overlap because of similar D_{aq}. (B) Ratio between the transverse dispersion coefficients of the different compounds in a wide range of seepage velocities.

Figure 3A shows the average values of the hydrodynamic transverse dispersion coefficient as function of velocity for all compounds. The plot clearly shows that the measured concentrations of the different solutes follow a distinct pattern. For the entire range of velocities investigated, the dispersion coefficients of bromide and oxygen are found to be significantly higher than the ones of fluorescein. This behavior is observed also for high velocities, for which mechanical dispersion should be the dominant process (corresponding to Péclet numbers $Pe > 5$ (*1*)). The theoretical ratio between the transverse dispersion coefficients for the three compounds analyzed in this study is plotted in Figure 3B as a function of the seepage velocity. It can be noticed that the biggest difference for the ratios of bromide/fluorescein and oxygen/fluorescein is expected at low velocities, where pore diffusion dominates transverse hydrodynamic dispersion. For the porous medium (average grain size of 0.625 mm) and the compounds (ratio between the high D_{aq} and the low D_{aq} of ~4) used in this study, the ratio between the transverse dispersion coefficients has a minimum (1.36) at 5.3 m/d. At higher velocities the ratios of transverse dispersion coefficients increase again with velocity. This clearly indicates that also the mechanical contribution to transverse dispersion is compound specific.

We interpreted these experimental results with the theoretical model proposed by Bear and Bachmat (*5*) because it already predicts a compound-dependent mechanical dispersion. An exponent β is introduced to account for the nonlinearities reported in previous studies (e.g., refs *10, 16, 29−35*) and it is assumed to be a fitting parameter in the following evaluation of the experimental data. In the model of Bear and Bachmat (*5*), it is assumed that the relative rate of growth of the volume occupied by a moving elementary mass of solute in a porous medium is linearly dependent on the characteristic velocities in

TABLE 1. Values for β, δ and the Reduced Chi-Squared Coefficient χ^2

compound	β [-]	δ [-]	χ^2
fluorescein	0.52	5.5	1.1
oxygen	0.49	5.4	1.3
bromide	0.53	5.8	1.1

the pore channels. The introduction of an exponent $\beta \neq 1$ is based on the hypothesis that the physical law for the volume growth rate may be nonlinear:

$$D_t \approx D_p + D_{aq}\left(\frac{Pe^2}{Pe + 2 + 4 \cdot \delta^2}\right)^{\beta} \tag{11}$$

Equation 11 has been used to evaluate the relationship between the measured D_t of each compound and the Péclet number. We fitted parameters β and δ by minimizing the squared residuals of the modeled and measured D_t values. The results are summarized in Table 1.

Average values of 0.5 and 5.5 were determined for the exponent (β) and for the ratio (δ) between the length and the width of a pore channel, respectively. The estimated uncertainty for these parameters is in the order of 10%. The obtained δ values were not significantly different from each other suggesting that the model is consistent with the experimental results. In fact, as mentioned above, the homogeneous porous medium used was the same in all experiments, and δ is supposed to be a property of the porous medium alone. A dependence of transverse dispersion on the velocity characterized with an exponent lower than one is in agreement with other experimental studies (*16, 31−33, 40*) and pore-scale modeling results (*23*). The continuous lines reported in Figure 3A represent the calculated values of D_t with the proposed parametric model (eq 12) for oxygen, bromide, and fluorescein.

Based on the outcome of the multitracer experiments, the relationship describing the hydrodynamic transverse dispersion in the given domain can be written as a function of the Pèclet number:

$$D_t = D_p + D_{aq}\sqrt{\left(\frac{Pe^2}{Pe + 2 + 4 \cdot (5.5)^2}\right)} = D_p + v\frac{d}{\sqrt{Pe + 123}} \tag{12}$$

At the limit of low flow velocities, pore diffusion is the dominant process. This holds for velocities up to ~1 m/d, depending on the compound and on the grain size of the matrix. Mechanical dispersion begins to be the dominant process at ~10 m/d. For compounds that diffuse faster, these velocities are higher, while for compounds with lower diffusion properties these velocities are smaller. The limit for very high flow velocities should be handled carefully, since Darcy's law holds only for Reynold's number Re = $(vd)/(N) \leq 1$ ($N \sim 1 \times 10^{-6}$ m^2/s being the kinematic viscosity of water) which, in the given domain, requires velocities <150 m/d.

The experimental results of this study and the new parametric model proposed (eq 12) were compared with literature data (*30, 32, 33, 42−44*), with grain sizes up to 1.25 mm and Péclet numbers up to 10^4. This range covers all cases of practical interest for flow in natural porous media. These literature data relate to flow-through experiments performed using tracer compounds with different aqueous diffusion coefficients, such as chloride ($D_{aq} = 2.03 \times 10^{-9}$ m^2/s), trichloroethene (TCE, $D_{aq} = 8.4 \times 10^{-10}$ m^2/s), and fluorescein ($D_{aq} = 4.8 \times 10^{-10}$ m^2/s). In Figure 4 the literature values of transverse dispersion coefficients and the measurements of this study are shown as a standard dimensionless plot. All data points appear to merge into a single line, as predicted by eq 12, since D_t is normalized by D_{aq} and plotted as a function of *Pe*. This graphical representation should be interpreted with caution since both values plotted (i.e., D_t/D_{aq} and *Pe*) are inversely proportional to D_{aq}, so that the compound-dependence of D_t is not obvious.

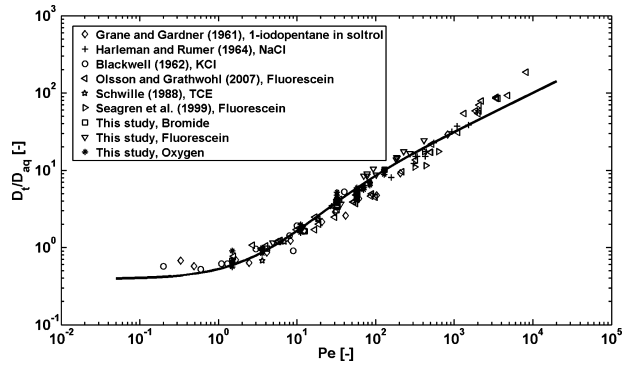

FIGURE 4. Dimensionless scatter plot of literature data (points) on transverse dispersion coefficients (D_t) compared with the new relationship (eq 12).

The comparison with previously published literature data allows generalizing the conclusions of this study to a wider range of compounds and grain sizes, even if the value of the parameters β and particularly δ may depend on the porous medium. The ratio between the length of a pore and its width depends on the geometry and the homogeneity of the grains that compose the porous material, while we suggest that the value for the exponent β should be kept constant. This value is in qualitative agreement with the outcomes of Klenk and Grathwohl (*29*) and the concept of incomplete mixing in the pores. The validity of eq 12 for each compound allows one to use the same empirical relation to describe transverse dispersion for different tracer solutions. This is an important outcome of this work, since standard power law fitting (e.g., refs *32, 33*) would lead to different coefficients for distinct compounds therefore preventing the formulation of a general empirical relationship.

The experimental evidence that transverse dispersion always depends on the compound-specific molecular diffusion coefficient implies that physical mixing is also compound-specific at any flow velocity. This outcome is relevant for mixing-controlled reactive transport in groundwater. The interpretation of experimental studies such as refs *16* and *45−47* could be improved by the introduction of a compound-dependent dispersion coefficient. Thus, a compound-specific description of transverse dispersion should be implemented in numerical codes simulating transport processes both at the laboratory and at the field scale. For instance, the efficient solution of transport and mixing-driven chemical and biochemical processes based on the distribution of mixing ratios (i.e., conservative species) recently introduced into the groundwater literature (*19, 48, 49*) may need to be reconsidered since this approach is based on the assumption that the dispersive behavior of all reaction partners is identical. Additional effort is required to quantitatively assess the influence of compound-dependent local-scale mixing in heterogeneous porous media. At the current state of knowledge, it is not clear to which extent compound-dependent behaviors will prevail in the macroscopic description of mixing at larger scales.

Acknowledgments

Support from the DFG (Deutsche Forschungsgemeinschaft) Research Group FOR 525 "Analysis and modeling of diffusion/

dispersion-limited reactions in porous media" (grants GR971/18-1, 18-3) is gratefully acknowledged.

Supporting Information Available

Additional information on the specific measurement techniques of the different tracer compounds and a comprehensive list of the multitracer experiments carried out in this study are provided. This material is available free of charge via the Internet at http://pubs.acs.org.

Literature Cited

(1) Bear, J. *Dynamics of Fluids in Porous Media*; Dover: New York, 1972.
(2) Slichter, C. Field measurement of the rate of movement of underground waters. *U.S.G.S. Tech. Rep. Water Supply* **1905**, 140.
(3) Bear, J. *Hydraulics of Groundwater*; Dover: New York, 1979.
(4) Taylor, G. I. Dispersion of soluble matter in solvent flowing slowly through a tube. *Proc. Royal Soc. A.* **1953**, *1137* (219), 186–203.
(5) Bear, J.; Bachmat, Y. A generalized theory on hydrodynamic dispersion in porous media. IASH Symposium on Artificial Recharge and Management of Aquifers, Haifa, Israel IASH, P.N., 1967, Vol. 72, pp 7−16.
(6) Bear, J. On the tensor form of dispersion in porous media. *J. Geophys. Res.* **1961**, *66*, 1185–1197.
(7) De Josselin de Jong, G. Longitudinal and transverse diffusion in granular deposits. *EOS Trans. AGU* **1958**, *39*, 67–74.
(8) Saffman, P. G. Dispersion due to molecular diffusion and macroscopic mixing in flow through a network of capillaries. *J. Fluid Mech.* **1960**, *2*, 194–208.
(9) Scheidegger, A. E. General theory of dispersion in porous media. *J. Geophys. Res.* **1961**, *66* (4), 3273–3278.
(10) Cirpka, O. A.; Frind, E. O.; Helming, R. Numerical simulation of biodegradation controlled by transverse mixing. *J. Contam. Hydrol.* **1999**, *40* (2), 159–182.
(11) Rolle, M.; Eberhardt, C.; Chiogna, G.; Cirpka, O. A.; Grathwohl, P. Enhancement of dilution and transverse reactive mixing in porous media: experiments and model-based interpretation. *J. Contam. Hydrol.* **2009**, *110*, 130–142.
(12) Liedl, R.; Valocchi, A. J.; Dietrich, P.; Grathwohl, P. Finiteness of steady state plumes. *Water Resour. Res.* **2005**, *31*, W12501; DOI 10.1029/2005WR004000.
(13) Kitanidis, P. K. The concept of dilution index. *Water Resour. Res.* **1994**, *30* (7), 2011–2026.
(14) Bauer, R. D.; Rolle, M.; Bauer, S.; Eberhardt, C.; Grathwohl, P.; Kolditz, O.; Meckenstock, R. U.; Griebler, C. Enhanced biodegradation by hydraulic heterogeneities in petroleum hydrocarbon plumes. *J. Contam. Hydrol.* **2009**, *105*, 56–68.
(15) Bauer, R. D.; Rolle, M.; Kürzinger, P.; Grathwohl, P.; Meckenstock, R. U.; Griebler, C. Two-dimensional flow-through microcosms: versatile test systems to study biodegradation processes in porous aquifers. *J. Hydrol.* **2009**, *369*, 284–295.
(16) Huang, W. E.; Oswald, S. E.; Lerner, D. N.; Smith, C. C.; Zheng, C. Dissolved oxygen imaging in a porous medium to investigate biodegradation in a plume with limited electron acceptor supply. *Environ. Sci. Technol.* **2003**, *37*, 1905–11.
(17) Anneser, B.; Einsiedl, F.; Meckenstock, R. U.; Richters, L.; Wisotzky, F.; Griebler, C. High-resolution monitoring of biogeochemical gradients in a tar oil-contaminated aquifer. *Appl. Geochem.* **2008**, *23*, 1715–1730.
(18) Prommer, H.; Anneser, B.; Rolle, M.; Einsiedl, F.; Griebler, C. Biogeochemical and isotopic gradients in a BTEX/PAH contaminant plume: model-based interpretation of a high-resolution field data set. *Environ. Sci. Technol.* **2009**, *43*, 8206–8212.
(19) Cirpka, O. A.; Valocchi, A. J. Two-dimensional concentration distribution for mixing-controlled bioreactive transport in steady state. *Adv. Water Resour.* **2007**, *30* (6−7), 1668–1679.
(20) Werth, C. J.; Cirpka, O. A.; Grathwohl, P. Enhanced mixing and reaction through flow focusing in heterogeneous porous media. *Water Resour. Res.* **2006**, *42*, W12414; DOI 10.1029/2005WR004511.
(21) Willingham, T. W.; Werth, C. J.; Valocchi, A. J. Evaluation of the effects of porous media structure on mixing-controlled reactions using pore-scale modeling and micromodel experiments. *Environ. Sci. Technol.* **2008**, *42* (9), 3185–3193; DOI 10.1021/es7022835.
(22) Acharya, R. C.; Valocchi, A. J.; Werth, C. J.; Willingham, T. W. Pore-scale simulation of dispersion and reaction along a transverse mixing zone in two-dimensional porous media. *Water Resour. Res.* **2007**, *43*, W10435; DOI 10.1029/2007WR005969.
(23) Bijeljic, B.; Blunt, M. J. Pore-scale modelling of transverse dispersion in porous media. *Water Resour. Res.* **2007**, *43*, W12S11; DOI 10.1029/2006WR005700.
(24) Boving, T.; Grathwohl, P. Tracer diffusion coefficients in sedimentary rocks correlation to porosity and hydraulic conductivity. *J. Contam. Hydrol.* **2001**, *53*, 85–100.
(25) Grathwohl, P. *Diffusion in Natural Porous Media: Contaminant Transport, Sorption/Desorption, and Dissolution Kinetics*; Kluwer Academic Publishers: Boston, MA, 1998.
(26) Saffman, P. G. A theory of dispersion in porous media. *J. Fluid Mech.* **1959**, *6* (3), 321–349.
(27) Freeze, R. A.; Cherry, J. A. *Groundwater*; Prentice-Hall International: NJ, 1979.
(28) Scheidegger, A. E. Statistical hydrodynamics in porous media. *J. Appl. Phys.* **1954**, *25*, 994–1001.
(29) Klenk, I. D.; Grathwohl, P. Transverse vertical dispersion in groundwater and the capillary fringe. *J. Contam. Hydrol.* **2002**, *58*, 111–128.
(30) Blackwell, R. J. Laboratory studies of microscopic dispersion phenomena. *Soc. Petrol. Engrs. J.* **1962**, *225*, 1–8.
(31) Carvalho, J. F. R.; Delgado, J. M. P. Q. Lateral dispersion in liquid flow through packed beds at Pe < 1400. *Am. Inst. Chem. eng.* **2000**, *46* (5), 1089–1095.
(32) Harleman, D.; Rumer, R. Longitudinal and lateral dispersion in an isotropic porous medium. *J. Fluid Mech.* **1963**, *16*, 385–394.
(33) Olsson, A.; Grathwohl, P. Transverse dispersion of non reactive tracers in porous media: A new nonlinear relationship to predict dispersion coefficients. *J. Contam. Hydrol.* **2007**, *92*, 149–161.
(34) Delgado, J. M. P. Q.; Carvalho, J. F. R. Measurement of the coefficient of transverse dispersion in flow through packed beds for a wide range of values of the Schmidt number. *Transp. Porous Media* **2001**, *44*, 165–180.
(35) Gutierrez, M. N. Aspects of transverse dispersion in porous media. Ph.D. Dissertation, Universiteit Utrecht, Netherland, 2009.
(36) Worch, E. Eine neue Gleichung zur Berechnung von Diffusionskoeffizenten gelöster Stoffe. *Vom Wasser* **1993**, *81*, 289–297.
(37) Atkins, P. W. *Physical Chemistry*, 4th ed; Oxford University Press: Oxford, UK, 1990.
(38) Cussler, E. L. *Diffusion, Mass Transfer in Fluid Systems*; Cambridge University Press: New York, 1984.
(39) Srinivasan, V.; Clement, T. P.; Lee, K. K. Domenico solution - is it valid? *Ground Water* **2007**, *45*, 136–146.
(40) Domenico, P. A.; Palciauskas, V. V. Alternative boundaries in solid waste management. *Ground Water* **1982**, *20* (3), 303–311.
(41) Delgado, J. A critical review of dispersion in packed beds. *Heat Mass Transf.* **2006**, *42* (4), 279–310.
(42) Grane, F. E.; Gardner, G. H. F. Measurements of transverse dispersion in granular media. *J. Chem. Eng. Data* **1961**, *6* (2), 283–287.
(43) Seagren, E. A.; Rittmann, B. E.; Valocchi, A. J. An experimental investigation of NAPL pool dissolution enhancement by flushing. *J. Contam. Hydrol.* **1999**, *37*, 111–137.
(44) Schwille, F. *Chlorinated solvents in porous and fractured media - model experiment*; Lewis Publishers: Boca Raton, Fl, 1988.
(45) Nambi, M. I.; Werth, J.; Sanford, R. A.; Valocchi, A. J. Pore-scale analysis of anaerobic halorespiring bacterial growth along the transverse mixing zone of an etched silicon pore network. *Environ. Sci. Technol.* **2003**, *37* (24), 5617–5624.
(46) Gramling, M. C.; Harvey, C. F.; Meigs, L. C. Reactive transport in porous media: a comparison of model prediction with laboratory visualization. *Environ. Sci. Technol.* **2002**, *36* (11), 2508–2514.
(47) Raje, S. D.; Kapoor, V. Experimental study of bimolecular reaction kinetics in porous media. *Environ. Sci. Technol.* **2000**, *34* (7), 1234–1239.
(48) De Simoni, M.; Sanchez-Vila, X.; Carrera, J.; Saaltink, M. W. A mixing ratios-based formulation for multicomponent reactive transport. *Water Resour. Res.* **2007**, *43*, W07419; doi:10.1029/2006WR005256.
(49) Saaltink, M. W.; Carrera, J.; Ayora, C. On the behaviour of approaches to simulate reactive transport. *J. Contam. Hydrol.* **2001**, *48*, 213–235.

ES9023964